

The empirical recurrence for OEIS A235738: recovering a conjecture that is not written down

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Abstract

OEIS A235738 records “Empirical recurrence of order 78”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 135$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 78, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 78 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235738 is “Number of $(n+1) \times (2+1)$ 0..2 arrays with the sum of each 2×2 subblock maximum and minimum lexicographically nondecreasing columnwise and nonincreasing rowwise.”. It has offset 1 and begins

576, 9710, 160075, 2423417, 36199157, 523547651, 7519969703, 106697433668, ...

The entry states:

Empirical recurrence of order 78 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:45:27, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 188$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 135$ of the 188, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 135$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 376$ exact terms of the model returns order 78 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	30	c_{21}	−14241910190348401	c_{41}	−2835082723410302620479	c_{61}	−43903590878
c_2	−93	c_{22}	−47369760447954986	c_{42}	−3193775798287571428021	c_{62}	−256097751517
c_3	−5041	c_{23}	111934464197623634	c_{43}	3663467818784358787409	c_{63}	91483083468
c_4	41820	c_{24}	301834991813494679	c_{44}	4174775113449328633882	c_{64}	574262390623
c_5	317491	c_{25}	−716439638873095922	c_{45}	−3977961542813534928339	c_{65}	16692999052
c_6	−4187436	c_{26}	−1604632725496596335	c_{46}	−4670787786027421544418	c_{66}	−10372321742
c_7	−9001636	c_{27}	3773682011029170231	c_{47}	3617325795506702954394	c_{67}	−5349347929
c_8	225800436	c_{28}	7175269257053929401	c_{48}	4456001534936415813494	c_{68}	148439402114
c_9	26799854	c_{29}	−16483756510953826513	c_{49}	−2741042026610009119176	c_{69}	9574261194
c_{10}	−7963347946	c_{30}	−27182731861462988783	c_{50}	−3610444207852816292716	c_{70}	−1643926307
c_{11}	7204550964	c_{31}	60055170497639773829	c_{51}	1718587544819900429614	c_{71}	−1121722293
c_{12}	200678039025	c_{32}	87795082725706251428	c_{52}	2474165876066551521667	c_{72}	1359223067
c_{13}	−299725713160	c_{33}	−183282302585075560105	c_{53}	−882448809393463325146	c_{73}	857753397
c_{14}	−3800588322475	c_{34}	−242992526118142287636	c_{54}	−1427778623408214867313	c_{74}	−790671727
c_{15}	7051210258469	c_{35}	470031431768946216000	c_{55}	365256517752101068263	c_{75}	−39159681
c_{16}	55874821234235	c_{36}	578427372413449361955	c_{56}	690606917111319396244	c_{76}	289150835
c_{17}	−116455816317529	c_{37}	−1015010686040545560644	c_{57}	−118628869493201166017	c_{77}	81395712
c_{18}	−652198805757791	c_{38}	−1186568149093356677149	c_{58}	−278537171389345090710	c_{78}	−5011144
c_{19}	1453921626086279	c_{39}	1847614515282784699927	c_{59}	28622548307451422784		
c_{20}	6145927729378202	c_{40}	2098125117928132839629	c_{60}	93110431931802082400		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 78, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $78 + 4$ terms removed returns order 78 as well, so $\deg q_{\min} = 78 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $78 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 78 that this sequence satisfies — and that family turns out to have exactly one member.

References

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